

Himel Das Chittagong, Bangladesh | +8801515222525 | himeldas077@gmail.com | [LinkedIn](#) | [Website](#) | [GitHub](#)

Summary:

MSc graduate in Applied Artificial Intelligence and Data Analytics with hands-on experience designing, developing, and deploying end-to-end machine learning and deep learning solutions. Proficient in Python, SQL, scikit-learn, TensorFlow, and FastAPI, with experience building production-ready ML pipelines, data preprocessing workflows, model evaluation, and cloud-deployed AI applications. Contributed to open-source machine learning projects and developed scalable AI solutions across predictive analytics, medical imaging, and retrieval-augmented generation (RAG) systems. Passionate about applying data-driven approaches to solve real-world problems and thrive in collaborative, fast-paced environments.

EDUCATION:

University of Bradford, Bradford, UK

Jan 2024 – Mar 2025

MSc in Applied Artificial Intelligence and Data Analytics

Relevant Courseworks: Machine Learning, Business Data Analytics, Artificial Intelligence, Big Data Visualization

University of Science and Technology Chittagong, Chattogram, Bangladesh

Jan 2015 – April 2019

BSc Engineering in Electrical and Electronic Engineering

SKILLS:

- **Programming languages:** Python, SQL, R, Python, MATLAB
- **ML/AI:** Scikit-learn, PyTorch, TensorFlow, Explainable AI (Grad-CAM, SHAP), LLMs, RAG, Hybrid Retrieval, Vector
- Databases, Predictive Modelling
- **Data & Visualisation:** Pandas, NumPy, Matplotlib, Power BI, Excel
- **Tools:** FastAPI, Streamlit, Docker Compose, GitHub Actions (CI/CD), pytest
- **Databases:** MySQL, ChromaDB, Supabase
- **Frontend:** Next.js
- **Soft Skills:** Problem-solving, Collaboration, Communication, Agile Mindset

PROJECTS:

Kidney Tumor Identification System – [TensorFlow | FastAPI | Streamlit | Docker | MLflow | DVC] [\[GitHub Repo\]](#)

- Developed an end-to-end deep learning system to classify kidney CT scans into four categories (Normal, Cyst, Tumor, and Stone) using an EfficientNetB4-based model, achieving **99.46% validation accuracy** and **98.38% test AUC-ROC**.
- Built a production-ready application with FastAPI and Streamlit, incorporating Grad-CAM visualizations, uncertainty estimation, PDF report generation, and containerized deployment using Docker.
- Applied MLOps practices including MLflow experiment tracking, DVC for data versioning, GitHub Actions for CI/CD, and automated data drift monitoring to support a reproducible machine learning workflow.

MediQuery – Role-Gated Medical RAG System – [FastAPI | Next.js | ChromaDB | BM25 | Docker | Groq LLM] [\[GitHub Repo\]](#)

- Developed a role-based medical knowledge assistant using Retrieval-Augmented Generation (RAG), combining BM25 and vector search with cross-encoder reranking to deliver context-aware, cited responses.
- Designed a secure retrieval pipeline with JWT-based role-based access control (RBAC), ensuring users could only access documents authorized for their roles before retrieval was performed.
- Built and containerized the full-stack application with FastAPI, Next.js, ChromaDB, and Docker, integrating conversation-aware query rewriting and query safety checks for reliable AI-assisted knowledge retrieval.

London Bike-Sharing Demand Prediction – [Python | scikit-learn | Streamlit] [\[GitHub Repo\]](#)

- Developed an end-to-end machine learning pipeline to predict hourly bike-sharing demand using environmental and temporal data, achieving an **R² score of 0.957** with a Random Forest model after pipeline optimization.
- Performed data preprocessing, feature engineering, model comparison, and evaluation to identify the most effective regression model for demand forecasting.
- Deployed the solution as an interactive Streamlit application with modular architecture, centralized YAML configuration, and an automated model training workflow for seamless cloud deployment.

CERTIFICATIONS / AWARDS:

- [Python: Beginner to Advanced For Data Professionals](#) Apr 2025
- [Get Job Ready: Power BI Data Analytics for All Levels](#) Oct 2023
- [SQL Beginner to Advanced For Data Professionals](#) Dec 2023
- [Complete the Google Data Analytics Professional Certification](#) Apr 2023

PROFESSIONAL DEVELOPMENT:

- **Information Visualization** July 2024
Aarhus University (via University of Bradford) | Funded by the Turning Scheme UK
- **Making Sense of Data: Handling, Processing, Visualizing, and Analyzing Data in R** Aug 2024
Aarhus University (via University of Bradford) | Funded by the Turning Scheme UK